Genomic & Pharmacogenomic Clinical Summary

Patient: Daniel Ehrle (Daniel_Ehrle) | **Date:** October 13, 2026

1. PRIORITY GENOMIC FINDINGS & CARRIER STATUS

GENE	VARIANT	ZYGOSITY	CLASSIFICATION	CLINICAL IMPACT & GUIDANCE
CBLIF	c.79+1G>A · rs147785187	Heterozygous	Pathogenic	Single-copy heterozygous carrier preserves intrinsic factor production. Check periodic serum B12 during routine wellness visits.
GJB2	p.Met34Thr · c.101T>C · rs35887622	Heterozygous	Pathogenic	Single-copy heterozygous carrier maintains normal adult hearing. Consider baseline hearing check; avoid unnecessary ototoxic aminoglycosides.
VDR	c.-1172A>G · rs4516035	Homozygous	Pathogenic	Modulates calcitriol receptor signaling. Check routine 25(OH)D at annual checkup to guide standard vitamin D3 supplementation.
F5	p.Arg534Gln · c.1601G>A · rs6025	Heterozygous	Modifier	Avoid systemic oral estrogen therapy (thromboembolism risk). Implement standard perioperative clot prophylaxis during major surgery.
PEX6	p.Val788Met · c.2362G>A · rs267608240	Heterozygous	Modifier	Single-copy heterozygous carrier preserves normal peroxisomal function. Standard annual liver and lipid wellness panels.
GJB3	p.Arg75Cys · c.223C>T · rs370476720	Heterozygous	Modifier	Single-copy heterozygous carrier maintains normal adult hearing. Consider baseline hearing check; avoid unnecessary ototoxic aminoglycosides.
SCN4A	p.His599Arg · c.1796A>G · rs187401185	Heterozygous	Modifier	Maintain normal electrolyte balance. Inform anesthesiology regarding depolarizing muscle relaxants prior to general surgery.

Prescribing Precautions & Annual Checkup Discussion Topics

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2. HIGH-PRIORITY MEDICATION & EXPOSURE PRECAUTIONS (CPIC / FDA)

GENE & DIPLOTYPE	PHENOTYPE	AFFECTED MEDICATIONS	PRESCRIBING ACTION & GUIDANCE
F5 p.Arg534Gln · c.1601G>A · rs6025 (rs6025 Heterozygous)	APC Resistance	⚠️ HIGH RISK · AVOID ESTROGENS **Systemic Estrogens / Oral Contraceptives**	**AVOID / THROMBOEMBOLISM RISK:** Avoid prothrombotic estrogen regimens (35-fold VTE / pulmonary embolism risk amplification); implement situational perioperative VTE prophylaxis (major surgery/immobilization).
CYP2C19 *1/*17 (I331V (rs3758581) A>G, expression (rs12248560) C>T)	Rapid Metabolizer	DOSE_DOWN Voriconazole, Clopidogrel, Mephenytoin, Methadone	**DOSE_DOWN:** Increased clearance and lower trough concentrations. Perform therapeutic drug monitoring (TDM) to avoid subtherapeutic levels.
CYP2C9 *1/*2 (R144C (rs1799853) C>T)	Intermediate Metabolizer	DOSE_DOWN Celecoxib (Celebrex), Phenytoin (Dilantin), Warfarin (Coumadin), Acenocoumarol	**DOSE_DOWN:** Moderate decrease in celecoxib clearance. Initiate therapy at lowest recommended dose and monitor for GI/cardiovascular adverse effects.
CYP3A5 *3/*3 (splice defect (rs776746) T>C)	Poor Metabolizer	DOSE_DOWN Carbamazepine, Clopidogrel, Cyclosporine, Efavirenz	**DOSE_DOWN:** Genotype TT is associated with increased dose of carbamazepine in people with Epilepsy as compared to genotype CC.
NAT2 *5/*5 (I114T (rs1801280) T>C)	Poor Metabolizer	CAUTION Isoniazid	**CAUTION:** NAT2 *5/*5 slow acetylator phenotype. Higher risk of isoniazid-induced hepatotoxicity and peripheral neuropathy. Monitor liver function tests (LFTs) and administer pyridoxine (Vitamin B6).

3. HIGH-LEVEL PREVENTIVE CARE & ANNUAL CHECKUP TOPICS

PREVENTIVE CARE TOPIC	TRIGGERING GENE & VARIANT	TIMING	CLINICAL CONTEXT & GUIDANCE
Periodic Vitamin B12 & Folate Check Routine Wellness Lab Panel	CBLIF c.79+1G>A · rs147785187 Heterozygous (Autosomal Recessive)	Routine Annual or Biennial Checkup	Single-copy heterozygous carrier preserves intrinsic factor production. Check periodic serum B12 during routine wellness visits.
Baseline Hearing Check & Ototoxic Medication Precaution Routine Preventive Care	GJB2 p.Met34Thr · c.101T>C · rs35887622 Heterozygous (Autosomal Recessive, Autosomal Dominant)	Baseline check; repeat if symptoms arise	Single-copy heterozygous carrier maintains normal adult hearing. Consider baseline hearing check; avoid unnecessary ototoxic aminoglycosides.
Routine Vitamin D & Calcium Check Annual Wellness Panel	VDR c.-1172A>G · rs4516035 Homozygous (Pathway Modifier)	Annual PCP Checkup	Modulates calcitriol receptor signaling. Check routine 25(OH)D at annual checkup to guide standard vitamin D3 supplementation.
Situational Clot Prevention (Surgery, Immobility, Estrogen Avoidance) Perioperative & Prescribing Precaution	F5 p.Arg534Gln · c.1601G>A · rs6025 Heterozygous (Autosomal recessive / AD)	Situational (Before surgery, long travel, or hormone therapy)	Avoid systemic oral estrogen therapy (thromboembolism risk). Implement standard perioperative clot prophylaxis during major surgery.
Routine Liver & Lipid Wellness Panel Standard Annual Physical Bloodwork	PEX6 p.Val788Met · c.2362G>A · rs267608240 Heterozygous (Autosomal recessive)	Annual PCP Checkup	Single-copy heterozygous carrier preserves normal peroxisomal function. Standard annual liver and lipid wellness panels.

Clinical Perspective, Evidence Context & EHR Chart Summary

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4. CLINICAL PERSPECTIVE: PROACTIVE CARE VS. AVOIDING OVER-INTERVENTION

✅ WHY THESE FINDINGS ARE USEFUL

- **Proactive Medication Safety:** Documenting pharmacogenomic diplotypes prevents avoidable drug side effects and subtherapeutic dosing before a prescription is started.
- **Routine Preventive Care:** High-level wellness topics integrate naturally into the annual primary care physical.
- **Specialist Coordination:** Sharing finding-specific precautions supports informed care without unnecessary medicalization.

🛑 WHY AGGRESSIVE INTERVENTION IS NOT NEEDED

- **Single-Copy Carrier Reassurance:** Single-copy recessive carriers retain a functional gene copy and maintain normal baseline health.
- **Absence of Biallelic Recessive Disease:** Zero two-copy pathogenic recessive disease genotypes are present in this profile.
- **Pathway Modifiers Are Not Diagnoses:** Common physiological modifiers should never trigger unprovoked medication changes or invasive testing.

5. CLINICAL EVIDENCE CONTEXT BY FINDING CATEGORY

FINDING CATEGORY	EVIDENCE TIER	CLINICAL INTERPRETATION BASIS
Primary Carrier & Susceptibility Findings (CBLIF, GJB2, VDR)	High	Established clinical curation records (ClinVar / OMIM) with clear inheritance context.
Pharmacogenomic Diplotypes (CYP2C19, CYP2C9, CYP3A5)	High	CPIC / DPWG clinical dosing guidelines with established star-allele assignments.
Physiological & Pathway Modifiers	Moderate	Population-level associations guiding general wellness discussions rather than diagnostic labeling.

6. COPY-PASTE PHYSICIAN HANDOFF / EHR CHART SUMMARY

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CLINICAL GENOMICS & PHARMACOGENOMICS SUMMARY (Daniel Ehrle | ID: Daniel_Ehrle):
- PRIORITY GENOMIC & CARRIER FINDINGS: CBLIF (c.79+1G>A · rs147785187, Heterozygous); GJB2 (p.Met34Thr · c.101T>C · rs35887622, Heterozygous); VDR (c.-1172A>G · rs4516035, Homozygous); F5 (p.Arg534Gln · c.1601G>A · rs6025, Heterozygous); PEX6 (p.Val788Met · c.2362G>A · rs267608240, Heterozygous). Single-copy recessive carrier findings retain one functional gene copy and do not cause primary recessive disease.
- PHARMACOGENOMIC PRECAUTIONS (CPIC / FDA): Follow standard CPIC dosing. Review pharmacogenomic diplotypes prior to prescribing substrate medications.
- PREVENTIVE CARE PLAN: Discuss high-level wellness and screening topics (Section 3) during routine primary care checkups.
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